ACT: an AI assistant for system administrators

5W's | Information brief | ACT on its own (no AAP)

ACT 0.6.25 (Linux and Windows) · October 2026

BLUF. ACT (Agentic Command Tool) is an **AI assistant for the command line**. An administrator types a problem in plain English ("nginx will not start, find out why"); ACT uses our GenAI.mil models to plan, runs the checks, reads the real output, and reports the **root cause** and the **fix**. It changes nothing without the administrator's **confirmation**, and it does not call a job done until a check **proves it worked**. One file each for Linux and Windows; nothing to install. **Ready for a pilot** with a few administrators.

WHO

uses, owns, oversees

- **Uses:** Linux and Windows system administrators, at their own terminal.
- **Owns:** the systems team: versions, settings, and the rules for how it may be used.
- **Oversees:** the ISSO decides what data may go to the GenAI service.
- **Relies on:** GenAI.mil, already in place. No new servers.

WHAT

an AI assistant at the command line

- Turns a plain-English task into a **plan**, runs the commands, reads the output, and reports **ROOT CAUSE, EVIDENCE, FIX, CONFIDENCE**.
- **Read-only checks run on their own; every change is shown and asks first.**
- Edits files safely (backup and diff) and **verifies** every change before it reports done.
- Not a sandbox: it acts with the permissions of the person running it.

WHEN

on demand, while an administrator works

- **Only when an administrator starts it.** Nothing runs in the background or on a schedule.
- Outages and errors, log analysis, configuration checks, checks after patching, and routine tasks ("which services failed since boot?").
- Optional **best-answer mode** for hard problems: every available model proposes a plan and the best one is used.
- **Now:** ACT 0.6.25 is released for Linux and Windows.

WHERE

what runs where

- **On the server the administrator is working on**, over their usual SSH (for example SecureCRT) or Windows session.
- **Linux:** one Python script (RHEL 8/9, Python 3.8+). **Windows Server:** one PowerShell script (5.1 or 7). No agent, no service, no open ports.
- **Model:** GenAI.mil over HTTPS from that server.
- **Key:** in the administrator's own settings file, readable only by them.

WHY

the payoff

- The first diagnostic pass (status, logs, config, disk) takes **minutes instead of an hour**; administrators spend their time on decisions.
- **The same careful method every time**, and junior administrators can see each step and learn from it.
- **Safer than copying commands from a chat window**: each command is checked, secrets are removed, and changes are confirmed and verified.
- Uses GenAI services we already pay for.

SAFEGUARDS

how it stays safe

- **Asks before every change.** Only proven read-only commands run without asking.
- Destructive commands (mass deletes, disk tools, firewall, reboot) always need **explicit confirmation**.
- **Passwords, tokens and keys are removed; host names, IP addresses and accounts are pseudonymized** (replaced with placeholders) before anything is sent to the model.
- Optional **audit log** of every step. **Recommended rule:** no hands-off "auto" mode on production.

Decisions requested

1. Approve a **pilot**: a few named administrators use ACT in its default ask-before-change mode, first on non-production servers.
2. **ISSO concurrence** on sending command output and log lines to GenAI.mil, with passwords, tokens and keys removed and host names, IP addresses and user names pseudonymized first.
3. Set the **usage rule**: default mode only on production (no "auto" mode), and name an **owner** for ACT versions and settings.

Source: ACT-Linux and ACT-Windows 0.6.25 documentation. Using ACT from Ansible Automation Platform is covered in a separate brief.

Backup: how it works, safeguards, status

ACT | 5W's

ACT 0.6.25 (Linux and Windows) · October 2026

How one task flows

1. Ask
administrator types the task in plain English

→

2. Plan
GenAI proposes steps; ACT keeps it on track

→

3. Investigate
read-only commands run; real output is read

→

4. Confirm
any change is shown; the administrator decides

→

5. Verify + report
a check proves the fix; root cause and evidence

The model never touches the server directly: it proposes one action at a time, and ACT decides whether that action may run.

Mode	What runs without asking	Use
Default (interactive)	Proven read-only commands only. **Everything else asks.**	Everyday use by an administrator. Recommended on production.
Auto (opt-in)	Almost everything; only catastrophic commands (mass deletes, disk destruction, firewall lockout, reboot) still ask.	Trusted, non-production work only. Recommended: not on production.
Non-interactive	Read-only commands; anything else is refused and reported as a proposed fix.	Scripts and automation (see the AAP brief).

Topic	Status / risk and mitigation
Testing to date	Released for Linux and Windows. Automated test suites pass on Python 3.8, 3.11 and 3.13 and on PowerShell 5.1 and 7. It already runs against GenAI.mil from AAP; a reply cut off on the way is asked for again (0.6.25).
Model can be wrong	The administrator confirms every change, and ACT refuses to report done until a check proves the fix. Diagnoses come with evidence.
Runs with the user's rights	Not a sandbox: ACT can do what the person running it can do. Use normal accounts; elevate only when needed.
Data sent to GenAI	The task and the output of the commands ACT runs. Passwords, tokens and keys are removed. IP addresses, fully qualified names, the server's own names, names listed for it, local accounts and e-mail addresses are **pseudonymized** (replaced with consistent placeholders); ACT puts the real values back only on the server, to run commands. A short name ACT cannot recognize may still be sent. In ACT since 0.6.18. Needs ISSO concurrence.
Prompt injection	A crafted log line could try to steer the model. Output is treated as untrusted data, and changes still need confirmation (the reason for no auto mode).
Network	Each server needs HTTPS access to the model endpoint.

Rollout: each phase moves on only when its exit criteria are met

1. Pilot (now)
A few named administrators, default mode, non-production servers.
Exit: diagnoses judged useful on real problems; data sent is acceptable.

2. Production, default mode
More administrators; production allowed with ask-before-change.
Exit: no unintended changes; the team relies on it for first-pass troubleshooting.

3. Scripts and automation
Diagnose-only runs from scripts or AAP (separate brief).
Exit: reports are used; any automatic fix is narrow, approved and verified.

Ongoing
Owner reviews versions, settings and usage rules.
Check: audit logs spot-checked; updates tested before rollout.

Source: ACT-Linux and ACT-Windows 0.6.25 documentation. Using ACT from Ansible Automation Platform is covered in a separate brief.